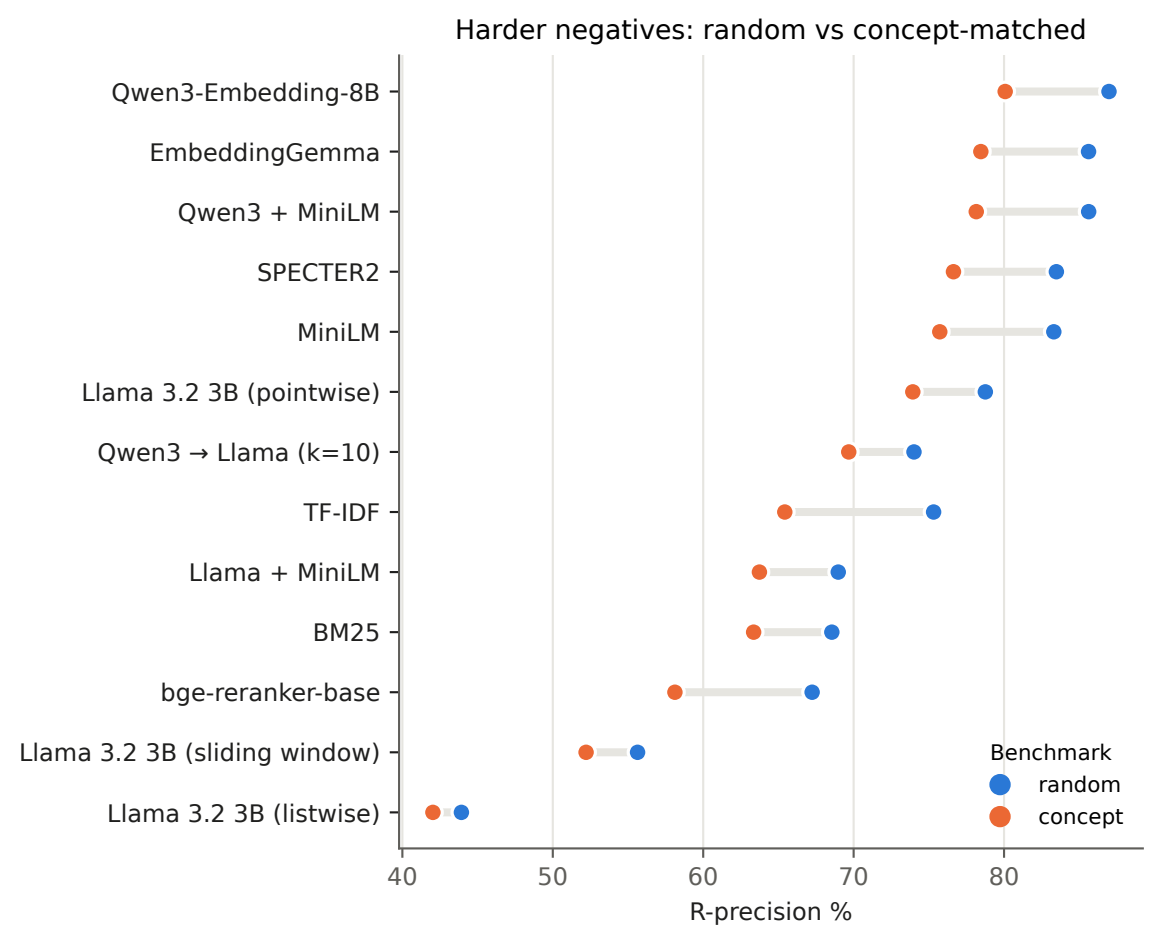
Harder negatives: random vs concept-matched
Qwen3-Embedding-8B
EmbeddingGemma
Qwen3 + MiniLM
SPECTER2
MiniLM
Llama 3.2 3B (pointwise)
Qwen3 → Llama (k=10)
TF-IDF
Llama + MiniLM
BM25
bge-reranker-base
Llama 3.2 3B (sliding window)
Llama 3.2 3B (listwise)
Benchmark
random
concept
40
50
60
70
80
R-precision %